

Hankel-Structured Gerchberg–Saxton Channel Estimation for Rydberg Atomic MIMO Receivers

Author Name

Department of Electrical and Electronics Engineering
Birla Institute of Technology and Science, Pilani
Email: f20220519@pilani.bits-pilani.ac.in

Abstract—Rydberg atomic quantum receivers measure only the *magnitude* of the incident field superposed on a local reference, making multi-user channel estimation a biased phase-retrieval rather than a linear-Gaussian problem. The known solvers—biased Gerchberg–Saxton (GS) and its expectation–maximisation refinement EM-GS—treat the channel as unstructured, and their normalised mean-square error (NMSE) consequently does not improve as the atomic array grows. We observe that on a uniform linear array each propagation path contributes exactly one spatial complex exponential, so the Hankel lifting of every channel column has rank at most the path count. Interleaving a Cadzow low-rank Hankel projection with every exact EM-GS update yields HS-GS; the model order comes from a held-out pilot residual, no oracle path count is used, and the magnitude observation is never linearised. We also derive unconstrained and geometry-constrained Cramér–Rao bounds for the exact Rician likelihood. Over 3.6×10^4 paired trials, and under the tested sparse geometric multipath model with i.i.d. uniform angles of arrival and equal per-path power, HS-GS improves on EM-GS by a mean of 2.85 dB at $N = 32$. Plotting the two error curves separately shows why: EM-GS is flat in array size to within 0.04 dB, so HS-GS is the only estimator whose NMSE improves as the aperture grows, and the gap widens monotonically ($0.03 \rightarrow 0.81 \rightarrow 2.45$ dB at $N = 8/16/32$). The benefit is conditional rather than general: at $N = 8$ the projection helps by 0.74 dB at -10 dB SNR but *hurts* by up to 0.40 dB above 0 dB. A controlled path-count experiment confirms the mechanism: the gain decays monotonically from 7.13 dB at $L = 2$ to zero exactly as the path count reaches the Hankel rank ceiling. Robustness under clustered propagation is not established and is the decisive next test.

Index Terms—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, Cadzow projection, constrained Cramér–Rao bound.

I. INTRODUCTION

Rydberg atomic quantum receivers replace the RF front end with an ensemble of highly excited atoms read out optically, giving sensitivity and bandwidth flexibility unavailable to conventional antennas [1], [2], [3]. Cui *et al.* [1] showed that the resulting atomic MIMO receiver does not obey the linear transmission model: heterodyne readout against a local reference yields the *modulus* of the summed field, so recovery is a *biased phase-retrieval* (PR) problem [4], [5]. They adapted the Gerchberg–Saxton (GS) algorithm [6] to this setting and proposed EM-GS, which replaces the hard modulus substitution with the exact posterior mean of the phase factor under the Rician law.

Both solvers treat $\mathbf{G} \in \mathbb{C}^{N \times K}$ as $2NK$ free real parameters. Under the magnitude model the Fisher information is block

diagonal across receive elements (Section V), so estimation separates into N independent per-row problems: enlarging the array adds proportionally as many unknowns as measurements and leaves the *normalised* error unchanged. Our simulations confirm this—EM-GS varies by only 0.012 dB across $N \in \{8, 16, 32\}$. Any improvement scaling with array size must therefore come from structure coupling the rows.

Such structure exists whenever the array is a ULA and propagation is via a small number of resolvable paths: each path imprints one spatial complex exponential across the aperture. This is the sparsity that grid-based and subspace estimators exploit in conventional MIMO, but it has not been used inside a magnitude-only atomic receiver, where the non-linearity forbids the usual linear least-squares machinery.

Contributions. We show that the Hankel lifting of each ULA channel column has rank at most L_k , turning path sparsity into a grid-free low-rank constraint requiring no angle estimation (Section III); we propose HS-GS, interleaving a Cadzow projection with every exact EM-GS update at an order chosen from a held-out pilot residual rather than from oracle knowledge (Section IV); we derive the unconstrained and geometry-constrained CRLBs for this Rician likelihood, correcting a rank-2 misconception of the per-measurement Fisher information (Section V); and we isolate *when* the method works with a controlled path-count experiment, stating explicitly the model conditions under which the claims hold (Sections VII–VIII).

Notation. $(\cdot)^T$ and $(\cdot)^H$ are transpose and conjugate transpose; $|\cdot|$ on a matrix is elementwise modulus; $\odot$ is the Hadamard product; $\|\cdot\|_F$ is the Frobenius norm; I_ν is the order- ν modified Bessel function of the first kind.

II. SYSTEM MODEL

A. Magnitude-Only Measurement

K single-antenna users transmit to an N -element atomic ULA over P pilot symbols. Writing $\mathbf{G} \in \mathbb{C}^{N \times K}$ for the channel, $\mathbf{S} \in \mathbb{C}^{K \times P}$ for the pilots, $\mathbf{B} \in \mathbb{C}^{N \times P}$ for the local reference and $\mathbf{W} \in \mathbb{C}^{N \times P}$ for noise, the receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}| \in \mathbb{R}_{\geq 0}^{N \times P}, \quad (1)$$

with $\mathbf{W}$ i.i.d. $\mathcal{CN}(0, \sigma^2)$ added *inside* the modulus. The phase of $\mathbf{GS} + \mathbf{B}$ is unobservable; only the reference $\mathbf{B}$ makes $\mathbf{G}$ identifiable, through the cross term of $|x + b|^2 =$

TABLE I
SIMULATION PARAMETERS. ALL EXPERIMENTS USE THE EXACT
OBSERVATION (1); THE MODULUS IS NEVER LINEARISED.

Channel and array		Estimators	
N	8, 16, 32	t_0	50 (GS, EM-GS)
K	3	T	50 (HS-GS)
L_k	$\mathcal{U}\{3, \dots, 7\}$ i.i.d.	project every	1 iteration
$\theta_{\ell,k}$	$\mathcal{U}[-\pi/2, \pi/2]$	n_{cz}	4 sweeps
$\alpha_{\ell,k}$	$\mathcal{CN}(0, \beta_k/L_k)$, $\beta_k=1$	ν	0.3 held out
c	1	T_{sel}	20
spacing	$\lambda/2$	$\hat{L}$ grid	$1, \dots, r_{\text{max}}$
Pilots and operating point		Monte Carlo	
P	10, 30 (swept 6–40)	trials/pt	400 (1200 at 5 pts)
$\mathbf{S}$	$\mathcal{CN}(0, 1)$, redrawn	total	2.8×10^4
	full row rank, $P \geq 2K$	bootstrap	2000, paired
$\mathbf{B}$	constant, $\angle b = 0$		
SNR	–5 to 20 dB, (2)		
RSR	12 dB (swept 0–24)		

$|x|^2 + 2\Re\{xb^*\} + |b|^2$. Following [1], the reference is constant over (n, p) with zero phase, the signal-to-noise ratio fixes

$$\sigma^2 = K/\text{SNR}_{\text{lin}}, \quad (2)$$

and the reference-to-signal ratio (RSR) is defined against a *single* user, giving $|b| = \sqrt{\text{RSR}_{\text{lin}}}$. Because K users transmit simultaneously, a nominal RSR of 12 dB corresponds to 7.15 dB measured against the K -user total; all RSR values quoted here are the single-user ones.

B. Geometric ULA Channel

The channel is generated as sparse geometric multipath. With half-wavelength spacing the electrical angle of path ℓ of user k is $\psi_{\ell,k} = \pi \sin \theta_{\ell,k}$, and

$$g_{n,k} = c \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \quad n = 1, \dots, N, \quad (3)$$

with atomic gain $c = 1$ (so the electrical and atomic channels coincide), $\alpha_{\ell,k} \sim \mathcal{CN}(0, \beta_k/L_k)$ and $\beta_k = 1$.

The assumptions used in every experiment reported here are: sparse geometric multipath with $L_k \sim \mathcal{U}\{3, \dots, 7\}$ drawn i.i.d. per user per realisation; angles of arrival i.i.d. *uniform in θ* (a Kolmogorov–Smirnov test over 4465 samples gives $p = 0.150$ against uniform-in- θ , $p < 10^{-4}$ against uniform-in-sin θ); equal average per-path power, so no near–far effect; pilots i.i.d. $\mathcal{CN}(0, 1)$, redrawn each trial and rejection-sampled to full row rank with $P \geq 2K$, hence *not* orthogonal (largest off-diagonal $|\mathbf{S}\mathbf{S}^H|/P$ is 0.233); and flat, narrowband propagation. The channel is *not* a clustered 3GPP TR 38.901 or Saleh–Valenzuela model; Section VIII addresses the consequences. Table I collects the parameters.

C. Baseline Solvers

Both baselines exploit the fact that row n of (1) depends only on row n of $\mathbf{G}$, reducing (1) to N independent biased PR problems of dimension K . Biased GS [1] alternates a least-squares update with the hard modulus substitution $\hat{\mathbf{y}} = \mathbf{Z} \odot$

$e^{j\angle(\mathbf{GS}+\mathbf{B})}$. EM-GS instead uses the exact conditional mean of the phase factor under the Rician law, which introduces the Bessel ratio

$$R(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}, \quad \kappa = \frac{2}{\sigma^2} \mathbf{Z} \odot |\mathbf{GS} + \mathbf{B}|, \quad (4)$$

so the restored observation is shrunk by $R(\kappa) \in [0, 1]$ according to measurement reliability. As $\kappa \rightarrow \infty$, $R \rightarrow 1$ and EM-GS provably reduces to biased GS—a limit reached in practice at high RSR (Section VII-A).

III. HANKEL STRUCTURE OF THE ULA CHANNEL

Drop the user index and write one channel column as $g_n = \sum_{\ell=1}^L \alpha_{\ell} z_{\ell}^n$ with $z_{\ell} = e^{-j\psi_{\ell}}$. Define the Hankel lifting with pencil parameter p ,

$$\mathcal{H}(\mathbf{g}) = \begin{bmatrix} g_0 & g_1 & \cdots & g_p \\ g_1 & g_2 & \cdots & g_{p+1} \\ \vdots & & \ddots & \vdots \\ g_{N-p-1} & \cdots & & g_{N-1} \end{bmatrix} \in \mathbb{C}^{(N-p) \times (p+1)}. \quad (5)$$

Its (q, r) entry is g_{q+r} , so

$$[\mathcal{H}(\mathbf{g})]_{q,r} = \sum_{\ell=1}^L \alpha_{\ell} z_{\ell}^{q+r} = \sum_{\ell=1}^L \alpha_{\ell} z_{\ell}^q z_{\ell}^r, \quad (6)$$

which separates in q and r . With $\mathbf{v}_{\ell} = [1, z_{\ell}, \dots, z_{\ell}^{N-p-1}]^T$ and $\mathbf{w}_{\ell} = [1, z_{\ell}, \dots, z_{\ell}^p]^T$ this is exactly

$$\mathcal{H}(\mathbf{g}) = \sum_{\ell=1}^L \alpha_{\ell} \mathbf{v}_{\ell} \mathbf{w}_{\ell}^T, \quad (7)$$

a sum of L rank-one terms, hence

$$\boxed{\text{rank}(\mathcal{H}(\mathbf{g})) \leq L}. \quad (8)$$

Each propagation path creates one spatial complex exponential across the ULA and therefore contributes exactly one rank-one component to the Hankel lifting, converting path sparsity into low-rank matrix structure without discretising the angular domain. When the z_{ℓ} are distinct and L does not exceed the matrix dimensions, (8) holds with equality, and conversely a sequence whose Hankel matrix has rank L is a sum of L exponentials [7].

The constraint has a hard ceiling: since $\mathcal{H}(\mathbf{g})$ is $(N-p) \times (p+1)$ its rank cannot exceed $\min(N-p, p+1)$, maximised over p at

$$r_{\text{max}} = \max_{1 \leq p \leq N-1} \min(N-p, p+1) = \lceil N/2 \rceil, \quad (9)$$

so $r_{\text{max}} = 4, 8, 16$ for $N = 8, 16, 32$. If $L \geq r_{\text{max}}$ then $\text{rank} \leq L$ is satisfied by every vector and carries no information. This ceiling states precisely *when* the prior is informative, and Section VII-C tests it directly.

IV. PROPOSED HS-GS ESTIMATOR

A. Cadzow Projection

Let $\mathcal{S}_{\hat{L}}$ be the length- N sequences whose Hankel lifting has rank at most $\hat{L}$. Membership is the intersection of two sets with exact projectors: fixed-rank matrices (truncated SVD, Frobenius-optimal by Eckart–Young) and Hankel matrices (anti-diagonal averaging). Cadzow’s algorithm [7] alternates them: build $\mathbf{H} \leftarrow \mathcal{H}(\mathbf{g})$; take its SVD $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$; retain the $\hat{L}$ dominant components as $\mathbf{H}_{\text{lr}} = \mathbf{U}_{:,1:\hat{L}}\mathbf{\Sigma}_{1:\hat{L}}\mathbf{V}_{:,1:\hat{L}}^H$; restore Hankel structure by averaging each anti-diagonal, $\mathbf{g} \leftarrow \mathcal{H}^\dagger(\mathbf{H}_{\text{lr}})$; and repeat for $n_{\text{cz}} = 4$ sweeps. The intersection is a non-convex variety and four sweeps do not reach its exact nearest point, so $\mathcal{P}_{\hat{L}}$ is an *approximate* projection (Section VII-D quantifies the residual) and the iteration below carries no convergence guarantee. When $\hat{L} \geq r_{\text{max}}$ the routine returns its input unchanged, so HS-GS reproduces EM-GS bit for bit.

B. Interleaved Iteration and Order Selection

The projection is applied *inside* the exact iteration, not as post-processing:

$$\mathbf{G}^{(t)} = \mathcal{P}_{\hat{L}} \circ \mathcal{T}_{\text{EM-GS}}(\mathbf{G}^{(t-1)}), \quad t = 1, \dots, T, \quad (10)$$

with $\mathcal{P}_{\hat{L}}$ acting column-wise after *every* one of the $T = 50$ updates. This matters because EM-GS update quality depends on the current iterate through κ in (4): a structurally cleaner iterate yields a better phase estimate, hence a better next iterate. Algorithms 1–4 give the measurement update, the projection, the rank selector and their composition; the paragraph after Alg. 4 states precisely which lines separate HS-GS from the baseline.

HS-GS requires a target rank $\hat{L}$, and the true L_k is unknown to any practical receiver. The in-sample residual cannot supply it: constraining $\mathbf{G}$ to a smaller set can only worsen the fit to the pilots used for fitting, so that criterion always prefers the largest candidate. Splitting the pilot columns into a fitting set and a validation set of fraction $\nu = 0.3$ penalises over-modelling correctly, and every quantity involved is observable:

$$\hat{L} = \arg \min_{1 \leq L \leq r_{\text{max}}} \|\mathbf{Z}_{\mathcal{V}} - |\mathbf{G}_L \mathbf{S}_{\mathcal{V}} + \mathbf{B}_{\mathcal{V}}|\|_F^2, \quad (11)$$

where $\mathbf{G}_L$ is the HS-GS estimate at order L from the fitting columns only, run for $T_{\text{sel}} = 20$ iterations. **No oracle rank is used anywhere in this paper**; the selector returns a single scalar $\hat{L}$ applied to all K columns. The order matters in both directions: $\hat{L} < L$ truncates genuine channel components, introducing a bias that does not vanish as noise decreases, whereas $\hat{L} > L$ admits additional noise and weakens the prior. Our selector systematically under-selects—with true orders (7, 4, 6) a typical choice is $\hat{L} = 5$, and in Section VII-C the mean $\hat{L}$ reaches only 11.55 at $L = 16$.

Relation to the baseline. Deleting lines 5–9 of Alg. 4 yields baseline EM-GS exactly: both chain the same $\mathcal{T}_{\text{EM}}$ for the same T iterations from the same initialisation, with no early stopping in either. Setting $\hat{L} \geq r_{\text{max}}$ makes $\mathcal{P}_{\hat{L}}$ the identity (Alg. 2, line 4), so the two estimators then agree *bit for bit*; this is checked numerically in §VI.

Algorithm 1 EM-GS measurement update $\mathcal{T}_{\text{EM}}$ (one iteration)

Require: $\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; current iterate $\mathbf{G}$ (or $\emptyset$)

- 1: **if** $\mathbf{G} = \emptyset$ **then** $\triangleright$ spectral initialisation, first call only
 - 2: $\mathbf{G} \leftarrow$ leading singular vectors of the row-wise $\mathbf{Z}$ -weighted Gram matrix
 - 3: **end if**
 - 4: $\mathbf{\Lambda} \leftarrow \mathbf{G}\mathbf{S} + \mathbf{B}$ $\triangleright$ noiseless field estimate, $N \times P$
 - 5: $\kappa \leftarrow (2/\sigma^2) \mathbf{Z} \odot |\mathbf{\Lambda}|$
 - 6: $\mathbf{R} \leftarrow R(\kappa)$, $R(x) = I_1(x)/I_0(x)$ $\triangleright$ elementwise; see §II-C
 - 7: $\hat{\mathbf{Y}} \leftarrow \mathbf{R} \odot \mathbf{Z} \odot e^{j\angle \mathbf{\Lambda}}$ $\triangleright$ Rician posterior mean of the field
 - 8: **for** $n = 1$ **to** N **do** $\triangleright$ rows are independent
 - 9: $\mathbf{G}_{n,:} \leftarrow (\hat{\mathbf{Y}}_{n,:} - \mathbf{B}_{n,:})\mathbf{S}^H(\mathbf{S}\mathbf{S}^H)^{-1}$
 - 10: **end for**
 - 11: **return** $\mathbf{G}$
-

Algorithm 2 Cadzow projection $\mathcal{P}_{\hat{L}}$ of one channel column

Require: $\mathbf{g} \in \mathbb{C}^N$, target rank $\hat{L}$, sweeps n_{cz}

- 1: $p \leftarrow \arg \max_{1 \leq p \leq N-1} \min(N-p, p+1)$ $\triangleright$ pencil maximising the attainable rank
 - 2: $r_{\text{max}} \leftarrow \min(N-p, p+1) = \lceil N/2 \rceil$
 - 3: **if** $\hat{L} \geq r_{\text{max}}$ **then**
 - 4: **return** $\mathbf{g}$ $\triangleright$ constraint vacuous: exact no-op
 - 5: **end if**
 - 6: **for** $t = 1$ **to** n_{cz} **do**
 - 7: $\mathbf{H} \leftarrow \mathcal{H}(\mathbf{g})$, $\mathbf{H}_{q,r} = g_{q+r}$ $\triangleright (N-p) \times (p+1)$
 - 8: $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \leftarrow \text{svd}(\mathbf{H})$
 - 9: $\mathbf{H} \leftarrow \mathbf{U}_{:,1:\hat{L}}\mathbf{\Sigma}_{1:\hat{L}}\mathbf{V}_{:,1:\hat{L}}^H$ $\triangleright$ projector 1: rank, Eckart–Young optimal
 - 10: $g_m \leftarrow \frac{1}{|\mathcal{A}_m|} \sum_{(q,r) \in \mathcal{A}_m} \mathbf{H}_{q,r}$, $\mathcal{A}_m = \{(q,r) : q+r = m\}$ $\triangleright$ projector 2: Hankel, by anti-diagonal averaging
 - 11: **end for**
 - 12: **return** $\mathbf{g}$
-

C. Complexity

One EM-GS update costs $O(NPK)$ once the $K \times K$ Gram matrix is cached; the projection adds Kn_{cz} SVDs of

Algorithm 3 Held-out pilot rank selection (uses no ground truth)

Require: $\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; held-out fraction ν , inner iterations T_{sel}

- 1: $P_{\mathcal{V}} \leftarrow \max(1, \text{round}(\nu P))$
 - 2: $\mathcal{V} \leftarrow \{P - P_{\mathcal{V}} + 1, \dots, P\}$, $\mathcal{F} \leftarrow \{1, \dots, P - P_{\mathcal{V}}\}$ $\triangleright$ validation / fitting columns
 - 3: **for** $L = 1$ **to** r_{max} **do**
 - 4: $\mathbf{G}_L \leftarrow \text{HS-GS}(\mathbf{Z}_{:, \mathcal{F}}, \mathbf{S}_{:, \mathcal{F}}, \mathbf{B}_{:, \mathcal{F}}, \sigma^2; L, T_{\text{sel}})$ $\triangleright$ Alg. 4, fitting columns only
 - 5: $J(L) \leftarrow \|\mathbf{Z}_{:, \mathcal{V}} - |\mathbf{G}_L \mathbf{S}_{:, \mathcal{V}} + \mathbf{B}_{:, \mathcal{V}}|\|_F^2$ $\triangleright$ scored on columns never fitted
 - 6: **end for**
 - 7: **return** $\hat{L} = \arg \min_L J(L)$
-

Algorithm 4 HS-GS: EM-GS with an interleaved structural projection

Require: $\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2; T, n_{\text{cz}}, \nu, T_{\text{sel}}, \Pi$ (project every Π iterations)

- 1: $\hat{L} \leftarrow \text{Alg. 3}$ $\triangleright$ omitted when $\hat{L}$ is supplied, as in Alg. 3 line 4
- 2: $\text{active} \leftarrow (\hat{L} < r_{\text{max}}); \mathbf{G} \leftarrow \emptyset$
- 3: **for** $t = 1$ **to** T **do**
- 4: $\mathbf{G} \leftarrow \mathcal{T}_{\text{EM}}(\mathbf{G})$ $\triangleright$ Alg. 1 – *identical* to the baseline
- 5: **if** active **and** $t \bmod \Pi = 0$ **then**
- 6: **for** $k = 1$ **to** K **do**
- 7: $\mathbf{G}_{:,k} \leftarrow \mathcal{P}_{\hat{L}}(\mathbf{G}_{:,k})$ $\triangleright$ Alg. 2, column-wise
- 8: **end for**
- 9: **end if**
- 10: **end for**
- 11: **return** $\hat{\mathbf{G}}_{\text{HS}} = \mathbf{G}$

an $(N - p) \times (p + 1)$ matrix per iteration, i.e. $O(Kn_{\text{cz}}N^3)$, and order selection multiplies the estimation by the r_{max} candidates. Against a *chained* EM-GS baseline—the correct reference, since HS-GS issues T single-iteration calls costing $1.005\text{--}1.038\times$ in overhead alone—HS-GS at fixed order costs $1.49\times$, $1.32\times$ and $1.22\times$ at $N = 8, 16, 32$, the ratio *falling* with N because the exact step grows faster than the projection at these dimensions. Order selection dominates the full estimator, at 57.7%, 74.1% and 85.5% of its runtime; shrinking the candidate set or reusing iterates across candidates is therefore the obvious practical target.

V. PERFORMANCE BENCHMARK

A. Likelihood and Fisher Information

Conditional on $\mathbf{G}, \mathbf{S}, \mathbf{B}$, the noiseless field at (n, p) is a known complex constant $\lambda = (\mathbf{GS} + \mathbf{B})_{n,p}$ and the measurement $z = |\lambda + w|$ is Rice distributed:

$$p(z | \lambda) = \frac{2z}{\sigma^2} \exp\left(-\frac{z^2 + |\lambda|^2}{\sigma^2}\right) I_0\left(\frac{2z|\lambda|}{\sigma^2}\right). \quad (12)$$

Crucially (12) depends on λ *only through* $a = |\lambda|$: the phase does not appear. Differentiating and using $\frac{d}{dx} \log I_0(x) = R(x)$,

$$\frac{\partial \log p}{\partial a} = \frac{2}{\sigma^2} [zR(\kappa) - a], \quad \kappa = \frac{2za}{\sigma^2}, \quad (13)$$

so a single measurement carries the scalar information $\beta(a, \sigma^2) = \mathbb{E}[(\partial_a \log p)^2]$, and its contribution to the Fisher matrix in the real coordinates $\boldsymbol{\eta} = [\Re \text{vec } \mathbf{G}; \Im \text{vec } \mathbf{G}] \in \mathbb{R}^{2NK}$ is

$$\mathbf{J}_n = \sum_{p=1}^P 4\beta(a_{n,p}, \sigma^2) \mathbf{u}_{n,p} \mathbf{u}_{n,p}^T, \quad (14)$$

with $\mathbf{u}_{n,p} = \partial a_{n,p} / \partial \boldsymbol{\eta}_n$. **Each magnitude measurement therefore contributes information of rank one, not rank two.** Crediting both quadratures overstates the information and yields a bound too low by 0.17–4.64 dB across our 36 operating points. An identifiability check settles the form: with

$K = 3$ (six real unknowns) and $P = 3$ measurements the per-row problem is unidentifiable and (14) is correctly singular, whereas the rank-two construction claims full rank. Fig. 1 corroborates it—EM-GS sits within 0.05 dB of the rank-one bound at high SNR, and would appear to violate the rank-two one. Because row n of $\mathbf{GS}$ enters only measurement row n , $\mathbf{J}$ is block diagonal across receive elements, which is why the unconstrained bound—and any estimator attaining it—is flat in N under a normalised metric.

B. Constrained Bound on the Geometric Manifold

The unconstrained bound governs estimators that treat $\mathbf{G}$ as $2NK$ free parameters. HS-GS instead exploits (3), whose parameters are $\boldsymbol{\phi} = \{\psi_{\ell,k}, \Re \alpha_{\ell,k}, \Im \alpha_{\ell,k}\} \in \mathbb{R}^{3 \sum_k L_k}$ —about 45 real numbers, *independent of* N . With $\mathbf{D} = \partial \boldsymbol{\eta} / \partial \boldsymbol{\phi}$ the textbook form $\mathbf{D}(\mathbf{D}^T \mathbf{J} \mathbf{D})^{-1} \mathbf{D}^T$ is unusable here, because $\mathbf{D}$ is rank deficient: at $N = 8$ the mean numerical rank of $\mathbf{D}$ is 40.4 against $3 \sum_k L_k \approx 44.7$ structural parameters, and 28.1% of realisations have $3 \sum_k L_k > 2NK = 48$ outright, so $\boldsymbol{\phi} \mapsto \mathbf{G}$ is not injective. We therefore use the Gorman–Hero / Stoica–Ng tangent-space form [8], [9], with $\mathbf{U}$ an orthonormal basis of $\text{range}(\mathbf{D})$ obtained by SVD at relative tolerance 10^{-9} :

$$\text{CCRB} = \mathbf{U}(\mathbf{U}^T \mathbf{J} \mathbf{U})^{-1} \mathbf{U}^T, \quad (15)$$

which depends on the tangent *space* and is thus invariant to over-parameterisation. The worst conditioning of $\mathbf{U}^T \mathbf{J} \mathbf{U}$ over all operating points is then 80.1, at the degenerate point $P = 6$; elsewhere it is below 35. Both bounds are averaged over the same 400 channel realisations used for the estimator curves, so the comparison is paired on the channel draw.

C. Bias Caveat

Both bounds constrain the covariance of *unbiased* estimators, whereas

$$\text{MSE} = \text{variance} + \text{bias}^2. \quad (16)$$

All three estimators run to a fixed budget with the shrinking update (4), and HS-GS additionally truncates rank, removing signal energy; none is unbiased, so neither bound applies strictly. Two consequences govern how the bounds are read. The unconstrained CRLB is the appropriate reference for GS and EM-GS only: HS-GS uses prior information that bound does not model, so it may legitimately fall below it and must be judged against the CCRB. And the CCRB, itself an unbiased-estimation bound on the manifold, does not forbid a sufficiently biased structured estimator from falling below it either. Section VII-E reports both effects with the accompanying shrinkage measurements.

VI. SIMULATION SETUP

All results use the exact observation (1); the modulus is never linearised, which is verified per trial. NMSE is pooled as a ratio of sums,

$$\text{NMSE}_G = 10 \log_{10} \frac{\sum_t \|\hat{\mathbf{G}}_t - \mathbf{G}_t\|_F^2}{\sum_t \|\mathbf{G}_t\|_F^2}, \quad (17)$$

never as a mean of per-trial decibels: (17) estimates the ensemble quantity a CRLB bounds, whereas a mean-of-decibels is strictly smaller by Jensen's inequality. The factor is 10, not 20, since the arguments are energies. Within an operating point all estimators see bit-identical realisations, so the reported gain

$$\Delta_{\text{HS}} = 10 \log_{10} \frac{\sum_t \|\hat{\mathbf{G}}_t^{\text{EM}} - \mathbf{G}_t\|_F^2}{\sum_t \|\hat{\mathbf{G}}_t^{\text{HS}} - \mathbf{G}_t\|_F^2} \quad (18)$$

and its 95% paired bootstrap interval (2000 resamples) benefit from cancellation of the common channel draw; realisations are independent *across* operating points.

Two averaging domains, kept distinct. Within an operating point, across trials, (17) sums energies first and converts to decibels last: the average is taken in the *linear* domain. Across SNR, for the summary of Fig. 6, the mean is of the per-SNR decibel values, i.e. *after* conversion. The distinction matters: linear NMSE spans three orders of magnitude over this grid, so a linear-domain average across SNR is dominated by the lowest-SNR point, which carries $\approx 70\%$ of the sum while the top three SNRs together carry 0.72%. Such an average would be nearly blind to the high-SNR half of the sweep, which is precisely where the projection hurts at $N = 8$.

Trial budget. Cost is dominated by rank selection, which reruns the estimator once per candidate order, so one trial at $N = 32$ costs $\approx 8.5 \times$ one at $N = 8$; the budget is therefore tiered and every table reports its own per-point count. In the SNR sweep of Section VII-B the budget is 400 trials per point, extended to 1200 at the five points whose intervals initially straddled zero. Section VII rests on 2.8×10^4 trials for the SNR, pilot-length and path-count sweeps, plus 8.4×10^3 paired trials on the $N \times \text{SNR}$ grid of Figs. 5–6 (600/400/200 per point at $N = 8/16/32$).

Verification. The estimator comparison is reproduced by a standalone package sharing the model and both estimators with the runs reported here, which passes thirteen automated checks. Two are load-bearing for fairness and are executed rather than asserted: each trial supplies a single *read-only* world $(\mathbf{G}, \mathbf{S}, \mathbf{B}, \mathbf{W}, \mathbf{Z})$ to both estimators, so they cannot diverge in channel, pilots or noise; and with the projection disabled HS-GS reproduces EM-GS to $\max |\text{diff}| = 0$, so the projection is provably the only difference. A third confirms the structural premise of Section III directly: a noiseless L -path column has Hankel singular values $\sigma_{L+1}/\sigma_1 \approx 7 \times 10^{-16}$, i.e. numerical rank exactly L . Where the two sets of runs overlap they agree bit for bit on all 200 common trials, so the differing summary values quoted in Sections VII-B and VII-D reflect only their different averaging grids ($P \in \{10, 30\}$ over six SNRs versus $P = 30$ over seven, the latter extended to -10 dB), not a difference in data.

VII. RESULTS AND DISCUSSION

A. Main Result and Array-Size Scaling

We first fix the operating point. EM-GS improves on biased GS by 1.31 dB at $\text{RSR} = 0$ dB, but the advantage collapses to 0.113 dB at 12 dB and 0.002 dB at 24 dB as $R(\kappa) \rightarrow 1$

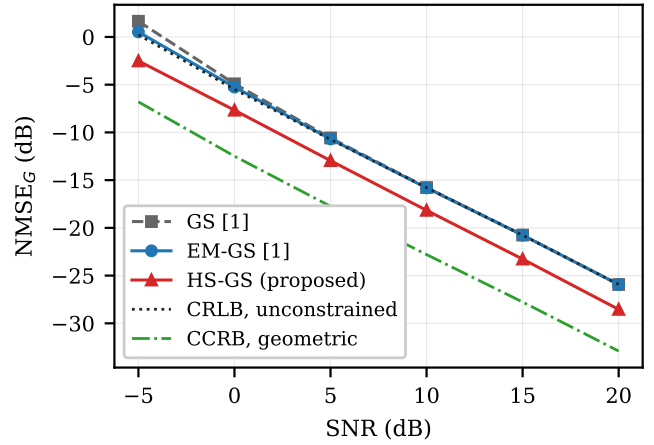


Fig. 1. NMSE versus SNR at $N = 32$, $P = 30$, $\text{RSR} = 12$ dB. EM-GS is indistinguishable from the unconstrained CRLB across the range, so the bound is tight for the unstructured problem. HS-GS lies between the two bounds, recovering part of the structural headroom identified by the geometric CCRB.

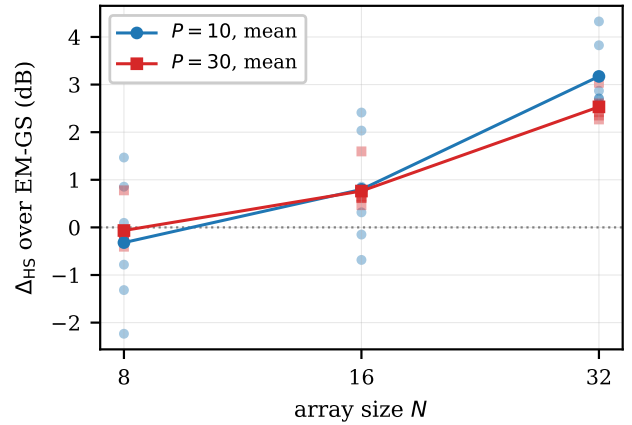


Fig. 2. Paired HS-GS gain over EM-GS versus array size. Faint markers are the six individual SNR operating points; solid lines are their mean. The gain grows monotonically with N ; at $N = 8$ the individual points straddle zero, so no single scalar summarises that array size.

and the two algorithms provably coincide. $\text{RSR} = 12$ dB, following the physical argument of [1], keeps the baselines separable and implies that the gains below are attributable to the structural projection rather than to the choice of exact step. Fig. 1 shows the principal comparison at $N = 32$, $P = 30$. EM-GS tracks the unconstrained CRLB to within 0.05 dB for $\text{SNR} \geq 5$ dB, confirming that it is close to efficient for the unstructured problem and that the corrected bound is credible. The geometric CCRB lies 0.87–9.98 dB below the unconstrained bound across all operating points, quantifying how much the structure is worth in principle. HS-GS occupies part of that gap, improving on EM-GS by 2.27 dB at 5 dB SNR and by 2.27–3.04 dB across the sweep.

Fig. 2 isolates the scaling. Averaged over the twelve operating points at each array size, the gain is -0.19 , $+0.78$ and $+2.85$ dB for $N = 8, 16, 32$, the per-trial win rate 33.5%, 74.0% and 95.3%, and the constraint active in 57.5%, 92.7%

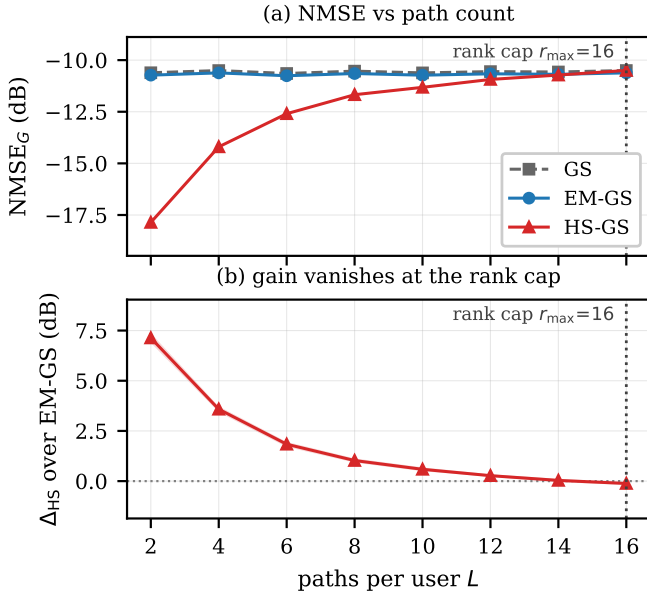


Fig. 3. Controlled path count ($N = 32$, $P = 30$, $\text{SNR} = 5$ dB, $\text{RSR} = 12$ dB, 400 trials per point, L fixed and identical across users). (a) The two baselines are flat in L —path count does not affect an estimator that ignores structure—which confirms the sweep is not confounded; HS-GS rises to meet them. (b) The paired gain decays monotonically to zero as L approaches the rank ceiling $r_{\max} = \lceil N/2 \rceil = 16$; the 95% paired bootstrap band is plotted but is narrower than the markers. The hypothesis producing this figure was recorded before the experiment was run.

and 99.6% of trials; the largest single-point gain is 4.33 dB ($N = 32$, $P = 10$, $\text{SNR} = -5$ dB). Under the same averaging EM-GS varies by only 0.012 dB across the three array sizes (-8.846 , -8.848 , -8.836 dB), exactly as the block-diagonal Fisher structure of Section V requires. The N -dependence therefore belongs to the structural prior, not to a degrading baseline.

The mechanism is the ratio of path count to rank ceiling (9). At $N = 8$, $r_{\max} = 4$ while $L_k \in \{3, \dots, 7\}$, so the selected order often reaches the cap, the projection is inactive in 43% of trials, and where active it truncates real components. At $N = 32$, $r_{\max} = 16 \gg L_k$ and the constraint is active in 99.6% of trials. We note that at $N = 8$ the sign of any scalar summary depends on the pooling—the per-point mean is -0.19 dB but energy-pooling all trials gives $+0.81$ dB—because HS-GS wins at low SNR and loses at high SNR there. That high-SNR loss is a bias-dominated floor: the truncation bias does not shrink with noise, so as variance falls the trade stops paying. At $N = 16$ and 32 every pooling agrees in sign.

B. Controlled Path Count: When and Why the Method Works

Every experiment above draws L_k at random, so none isolates the path count. Fig. 3 fixes $L_k = L$ identically across users and sweeps it to $r_{\max}$. The gain decays strictly monotonically,

$$7.13, 3.58, 1.84, 1.03, 0.59, 0.27, 0.04, -0.12 \text{ dB}$$

for $L = 2, 4, \dots, 16$, with the per-trial win rate falling from 100% to 45.3% and EM-GS flat to 0.140 dB across the sweep.

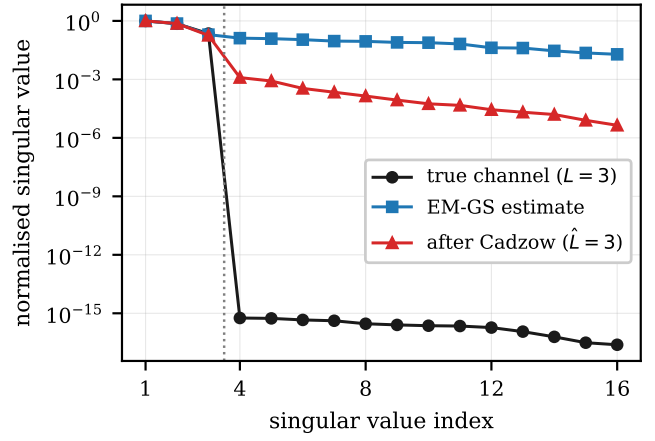


Fig. 4. Normalised Hankel singular values for one channel column, $N = 32$, $L = 3$, $\text{SNR} = 5$ dB. The true channel is exactly rank L ; the EM-GS estimate is full rank; Cadzow restores the cliff approximately, to $\sim 10^{-3}$ rather than to machine precision.

The interpretation is direct: when $L \ll r_{\max}$ the Hankel lifting carries substantial redundant structure, and projecting onto the low-rank Hankel set suppresses estimation error that does not share that structure. As $L \rightarrow r_{\max}$ the prior admits almost every vector and the benefit disappears.

Two details deserve emphasis. First, the null is reached at $L = 14$ (gain $+0.04$ dB, CI $[-0.05, +0.12]$), one grid point *before* the ceiling, and at $L = 16$ the gain is small but significantly negative (-0.12 dB, CI $[-0.195, -0.041]$). This is explained by the selector rather than by the theory: mean $\hat{L}$ reaches only 11.55 at $L = 16$, so the constraint is still active in 90.2% of trials and what it does there is discard genuine channel components—the $\hat{L} < L$ regime of Section IV-B. Second, this result rules out the most plausible rival explanation. If the improvement were generic denoising it would persist at large L , where the estimate is equally noisy; instead it vanishes and reverses exactly where the structural prior becomes vacuous.

C. Structural Diagnostic

Fig. 4 shows why the prior exists and what the projection achieves. The Hankel spectrum of the true channel falls to $\sim 10^{-16}$ immediately after index L : the low-rank property is exact, as (8) states. The EM-GS estimate has no such cliff—noise fills the whole spectrum, so the estimate is not a sum of L exponentials and the structural information is absent from it. After Cadzow a cliff reappears at the correct index but descends only to $\sim 10^{-3}$. That shortfall is the four-sweep projection being approximate, and it upper-bounds how much structure can be imposed—consistent with HS-GS attaining only part of the CCRB headroom.

D. Absolute NMSE of Both Estimators versus N

Fig. 2 plots the *difference* Δ_{HS} , which leaves open how each estimator behaves on its own. Figs. 5–6 plot the two NMSE curves themselves, over an SNR grid extended down to -10 dB, at $P = 30$.

Two facts separate cleanly. First, **EM-GS is flat in N** : its SNR-averaged NMSE is -10.321 , -10.337 and -10.301 dB at $N = 8, 16, 32$, a spread of 0.036 dB, and the same flatness holds inside every panel of Fig. 5 (for instance -25.90 , -25.95 , -25.87 dB at 20 dB SNR). Under a normalised metric, enlarging the array buys the unstructured estimator essentially nothing, exactly as the block-diagonal Fisher structure of Section V requires. Second, **HS-GS is the only curve that moves**: -10.350 , -11.149 , -12.753 dB. The gap therefore widens monotonically, $0.03 \rightarrow 0.81 \rightarrow 2.45$ dB, and the ordering holds at every SNR tested.

The $N = 8$ panel deserves its own sentence, because the SNR-averaged 0.03 dB conceals a sign change: HS-GS gains 0.744 dB at -10 dB SNR but loses 0.190 – 0.403 dB at 0 dB and above, with confidence intervals excluding zero in both directions. At that array size $r_{\max} = 4$ is comparable to $L_k \in \{3, \dots, 7\}$, so the projection is either inactive or truncating genuine channel components. $N = 8$ is a mixed result, not a small positive one.

E. Distance to the Bounds, and Crossings

HS-GS lies above the geometric CCRB at 34 of 36 operating points, by 0.32 – 8.85 dB, and the gap *grows* with N ($+0.32$ to $+1.41$ dB at $N = 8$; $+4.29$ to $+4.82$ dB at $N = 32$, $P = 30$). HS-GS therefore captures more structural gain in absolute terms as the array grows but a smaller share of what is available—the remaining headroom lies in the two weaknesses identified above, the shared scalar $\hat{L}$ and the approximate projection.

Curves do fall below the plotted bounds, and two situations must be separated. HS-GS lies below the *unconstrained* CRLB at 26 of 36 points, by up to 6.94 dB, but this is not a violation: that bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters, whereas HS-GS exploits a prior it does not model. The CCRB is HS-GS’s appropriate bound, and it is respected at 34 of 36 points. The remaining crossings are consistent with bias, as (16) permits. GS and EM-GS fall below the unconstrained CRLB at 18 and 19 of 36 points, but the excursions are strongly SNR-ordered: up to 2.34 and 2.62 dB at -5 dB, 1.05 and 1.16 dB at 0 dB, and at most 0.05 dB for $\text{SNR} \geq 15$ dB, within Monte Carlo error. The measured shrinkage coefficient $c = \langle \hat{\mathbf{G}}, \mathbf{G} \rangle / \|\mathbf{G}\|_F^2$ follows the same ordering: at $(N, P) = (32, 30)$, $c_{\text{EM}} = 0.944$ at -5 dB but 1.002 at $+5$ dB, where the crossings have vanished. HS-GS falls below the CCRB at exactly the two lowest SNRs of $(N, P) = (8, 10)$, precisely the points of strongest shrinkage: $c_{\text{HS}} = 0.561$ at -5 dB and 0.776 at 0 dB, against 0.976 at 20 dB where no crossing occurs. HS-GS shrinks more than the baselines at every SNR, since rank truncation discards signal energy on top of the shrinkage already in (4). We do not claim bias mathematically accounts for every crossing, only that the shrinkage measurements track them in magnitude and location and that no other mechanism was identified.

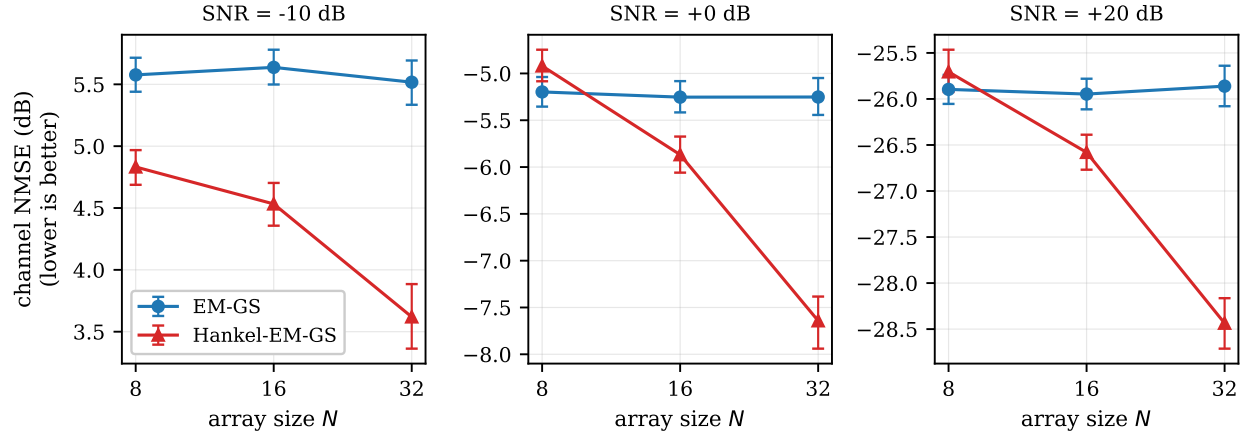
VIII. LIMITATIONS AND FUTURE WORK

All results here were produced under the i.i.d. uniform-AoA, equal-per-path-power, flat geometric model of Section II, so the defensible claim is model-conditional: *under the tested sparse geometric multipath model, exploiting the low-rank Hankel structure of the ULA response improves channel-estimation NMSE relative to EM-GS, by a mean of 2.85 dB at $N = 32$* . No result here establishes robustness under clustered propagation. The decisive next experiment is to repeat the principal comparison under a clustered 3GPP TR 38.901 generator [10]—a falsification test, not a confirmation exercise. Under clustering, rays within a cluster occupy a narrow angular window, so their z_ℓ are tightly packed and $\mathcal{H}(\mathbf{g})$ becomes only *numerically* low rank with a rapidly decaying spectrum rather than an exact cliff. Hard truncation, exactly right for the latter, is then a bias-inducing approximation, and the order selector may either adapt to the effective rank or chase the decaying tail. Since the $N = 8$ high-SNR degradation already traces to truncation bias, clustering could plausibly weaken or eliminate the gain.

Three further limitations are noted. A single scalar $\hat{L}$ is imposed on all K users despite their differing true orders; per-user selection would change only the selector and would isolate how much of the CCRB gap is order misspecification. The iteration (10) has no convergence guarantee—non-convex constraint set, approximate projection—and is defined by a fixed budget. Finally, all users have equal average power, so near–far effects are untested.

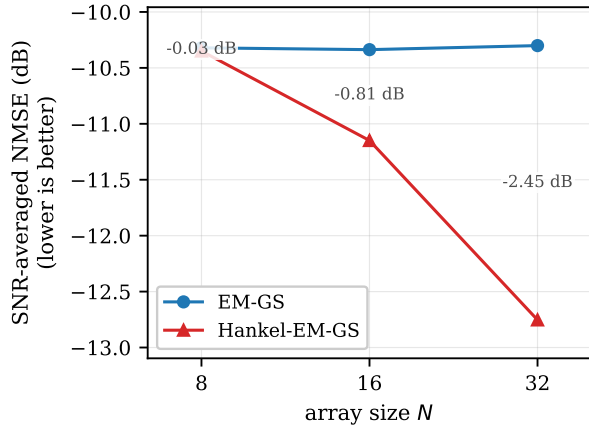
IX. CONCLUSION

The ULA response of a sparse geometric channel makes the Hankel lifting of each channel column rank-deficient, and this structure can be enforced inside an exact biased phase-retrieval iteration without linearising the magnitude observation and without estimating any angle. The resulting HS-GS estimator improves on EM-GS by a mean of 2.85 dB at $N = 32$ under the tested model. Plotting absolute NMSE rather than the difference makes the asymmetry explicit: the unstructured baselines are flat in array size, as the block-diagonal Fisher information requires, so HS-GS is the only estimator that converts a larger aperture into a lower normalised error. That the gap widens with N is an *empirical* finding over $N \in \{8, 16, 32\}$, consistent with the rank ceiling $\lceil N/2 \rceil$ growing past L_k but not proved to hold beyond the sizes simulated; the benefit is also conditional, being negative at $N = 8$ above 0 dB SNR. A controlled path-count experiment both confirms the mechanism and delimits it: the gain scales with the redundancy $r_{\max}/L$ and vanishes when the path count reaches the Hankel ceiling. Against a constrained CRLB derived for this likelihood, HS-GS attains part but not all of the available structural gain, localising the remaining headroom in per-user order selection and in the approximate projection. Validating the method under clustered propagation remains the essential next step.



$K=3$, $P=30$, $RSR=12$ dB, $L_k \sim \mathcal{U}\{3..7\}$; paired trials, identical channel/pilots/noise/initialisation per trial; error bars 95% bootstrap CI

Fig. 5. Absolute channel NMSE of both estimators versus array size at three SNR regimes ($K = 3$, $P = 30$, $RSR = 12$ dB, $L_k \sim \mathcal{U}\{3..7\}$; 600, 400 and 200 paired trials per point at $N = 8, 16, 32$). Lower is better. Error bars are 95% bootstrap intervals. EM-GS is flat in N in all three panels; HS-GS is the only curve that improves. Note the leftmost panel: at $N = 8$ the curves nearly touch, and at 0 and 20 dB HS-GS is *above* EM-GS.



mean of the seven per-SNR NMSE values (averaged AFTER the dB conversion); labels give Hankel – EM-GS

Fig. 6. SNR-averaged NMSE versus array size, the mean of the seven per-SNR values taken *after* conversion to decibels (see §VI). Labels give HS-GS minus EM-GS.

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